
The traveling
salesman
problem

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(2)

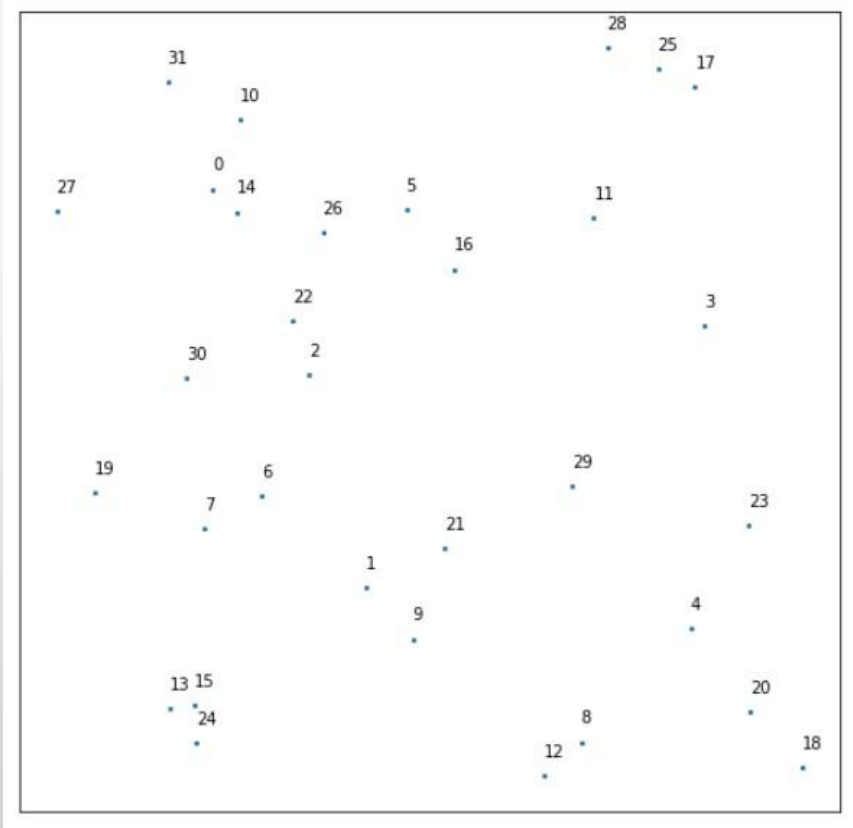
The traveling salesman problem

The traveling salesman problem

- A traveling salesman has to find the shortest closed tour between a set number of cities N
- Every **distances** between the cities are known
- Problem: (N-1)! possibilities
- Solution: use algorithm
  - Only quasi-optimal results

Our example

- Distribution of 32 randomly distributed points $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ with $x, y \in [0,2]$
- $\mathrm{d_{ij}} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$
- Symmetric matrix



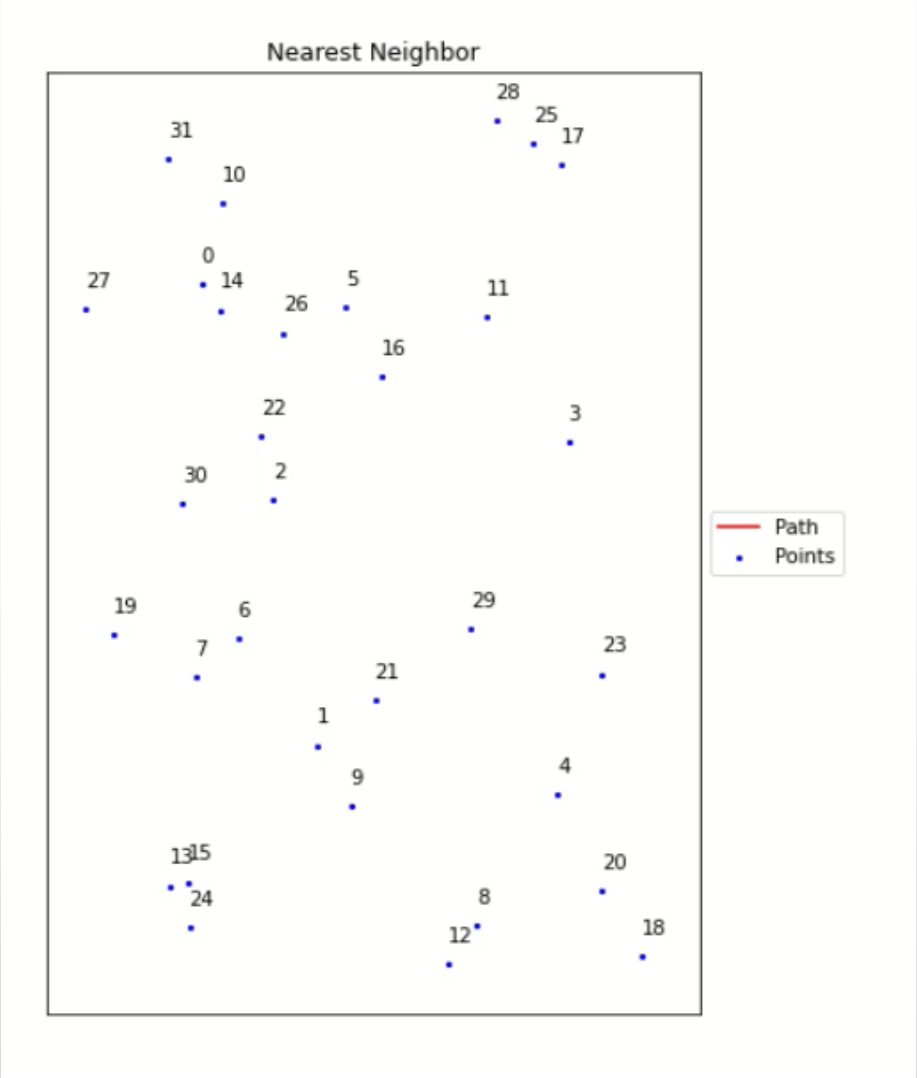
Greedy algorithm

Greedy Algorithm

- problem-solving heuristic of making the **locally** optimal choice at each stage
- Looking at three different approaches:
  - Nearest Neighbor
  - Double-sided nearest Neighbor
  - Nearest Insertion

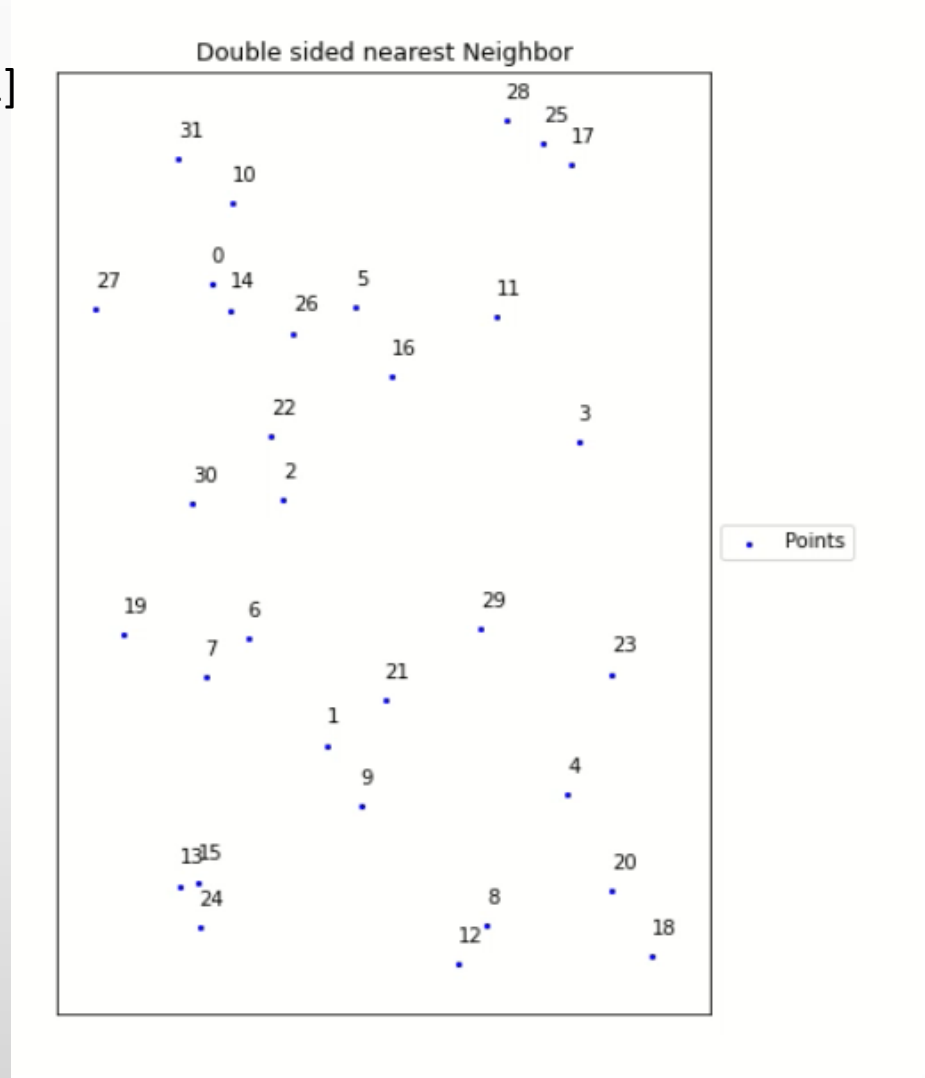
Nearest Neighbor [1]

- Start with a random start position
- Add always the cheapest edge to an unused vertex, until all vertices are visited once
- Add the starting vertex
- Do it for all start positions and choose the shortest path



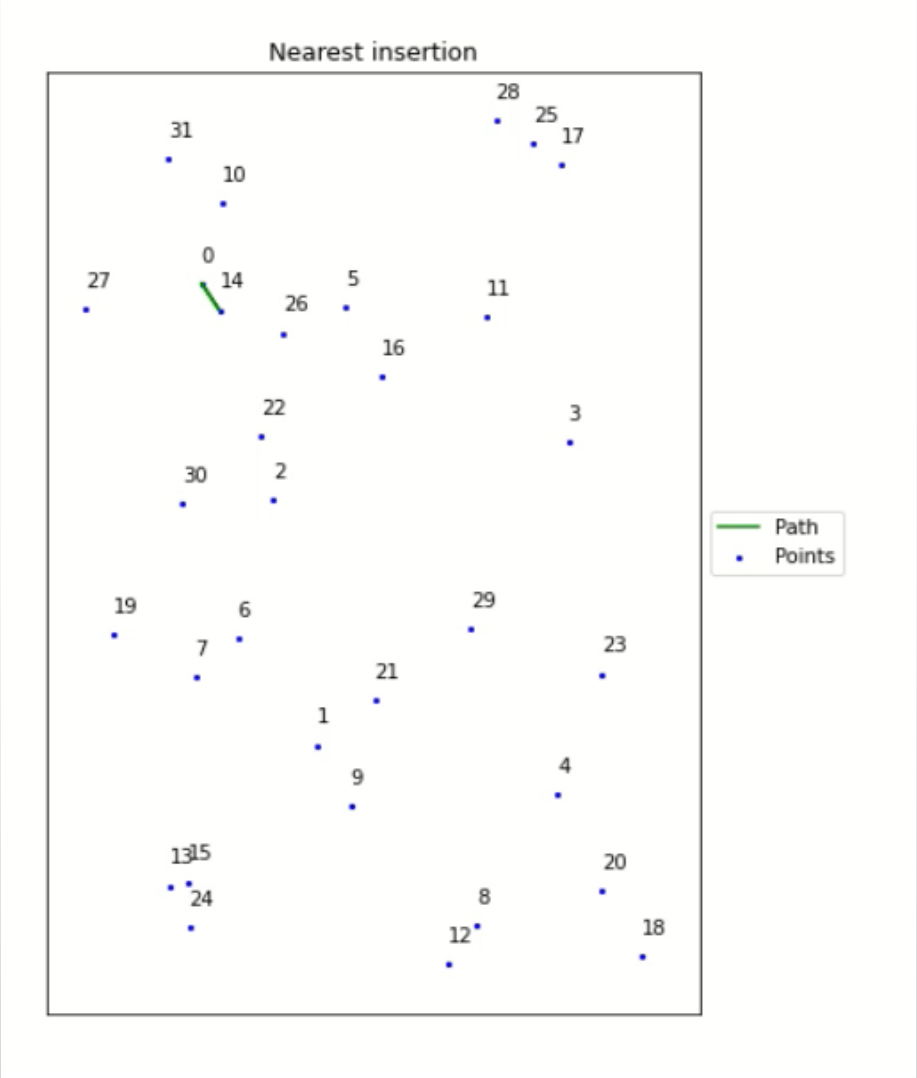
Double- sided Nearest Neighbor[1]

- Same algorithm as for nearest Neighbor, but go both ways
- Start with the shortest edge in one direction and second shortest in the other



Nearest Insertion [2]

- Start with a circle
- Add vertex, so that the circumference of the circle stays the lowest for all possibilities



Monte-Carlo Markov-Chain

MCMC Algorithm [3]

- Idea : Instead of creating a path using certain conditions, alter a previous path in a certain way.
- Code:
  - Start with random tour and calculate total distance ***D***
  - For a number of steps
    - Change tour with optimization method and calculate ***D****
    - If $x \sim Unif[0,1] < \exp\left(\frac{D-D^*}{T}\right)$ accept new tour else use old tour
    - ***D*** = ***D****
    - $T_{new} = \alpha \cdot T$

Optimization Method [3]

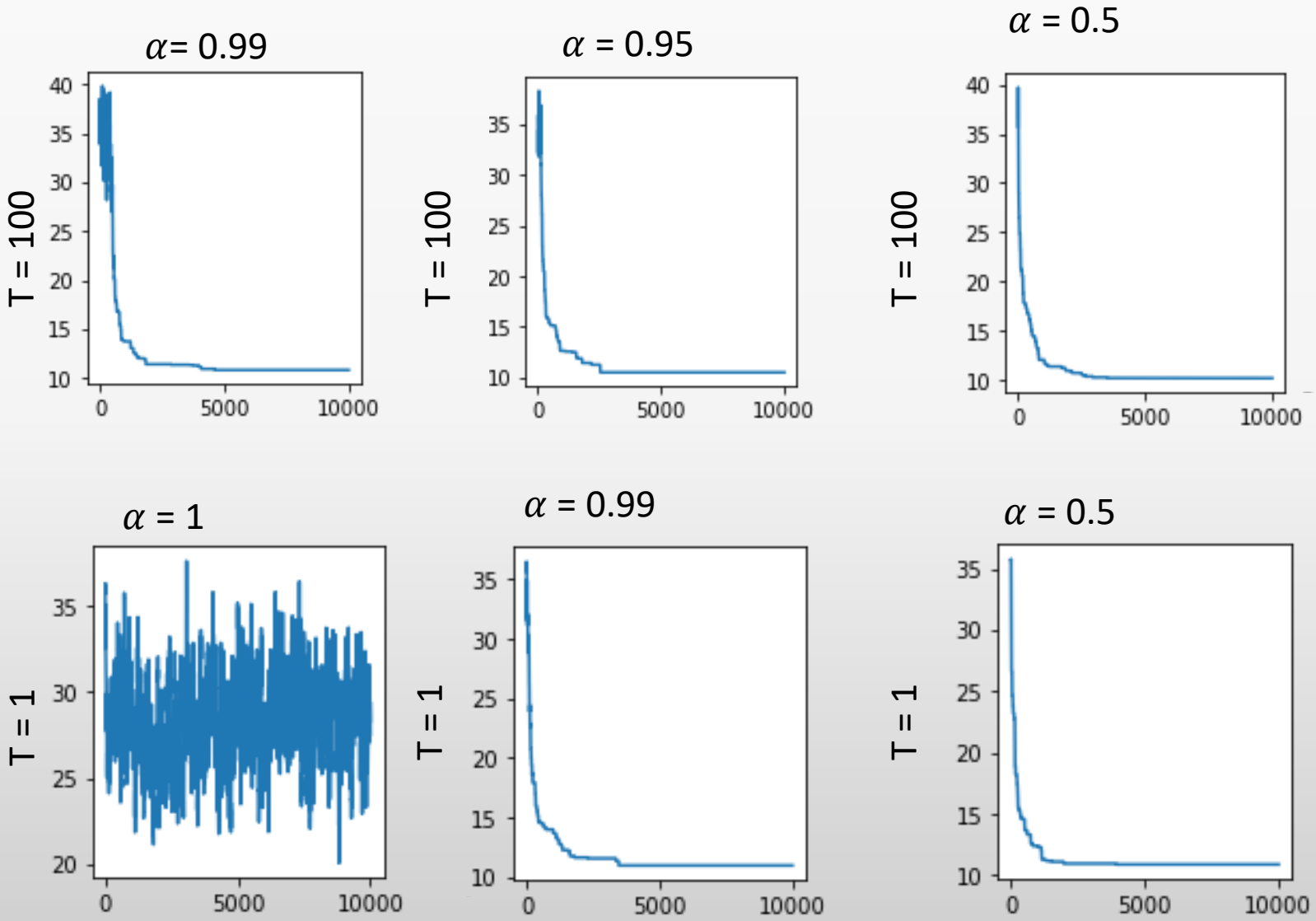
- Optimization method: randomly between 2-Opt and 3-Opt method
- 2-Opt method: deletes two edges thus creating 2 subtours and reconnects it with opposite edges
  - for each new proposed path reverses the direction of a part of the tour
- 3-Opt method: removes three edges, therefore breaking the tour into 3 separate sub-tours. Reconnecting by exchanging two sub-tours
  - exchanges two random parts of the tour without altering their directions

Role of T and α

- *T* and α serves as control parameter
- By lowering the *T* parameter, the system is transferred from a random high-cost to a feasible low-cost solution
- Often accept changes at the beginning , but later only accept cheaper paths
  - escape local minima at the beginning
  - trapped in a minima at the end -> Might not be global Minima

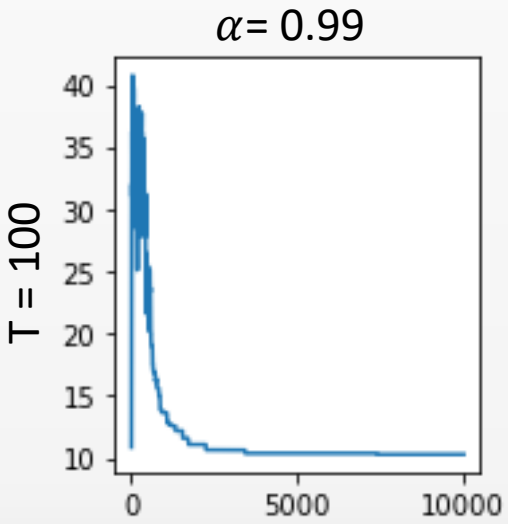
T and α - non optimal start

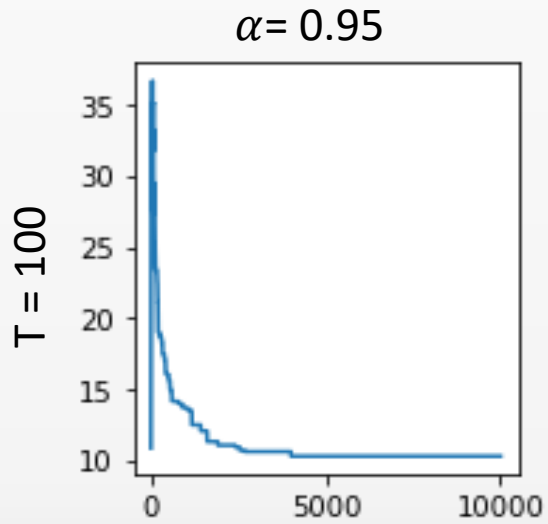
- Result for distance of different α and T with 10000 steps MCMC starting with 0,1,…,32
- We have $\exp\left(\frac{D-D^*}{T}\right)$ with $\max(D - D^*) \approx 2{,}5$

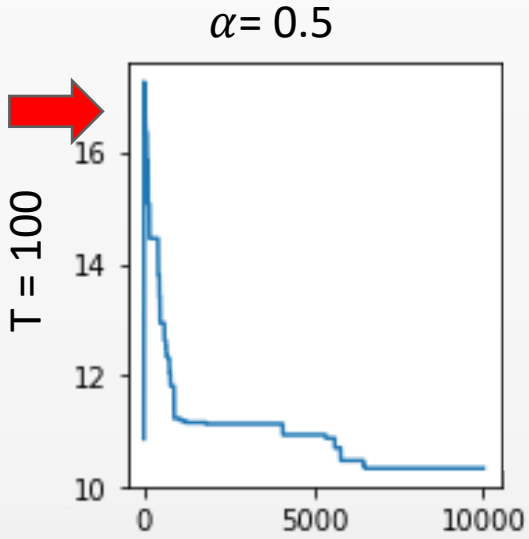


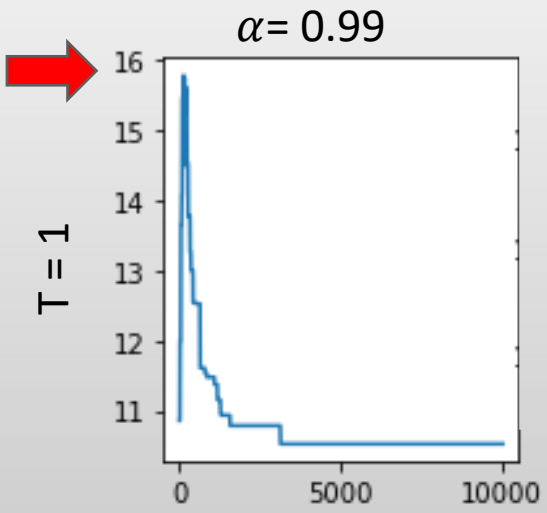
T and α – start in local minimum

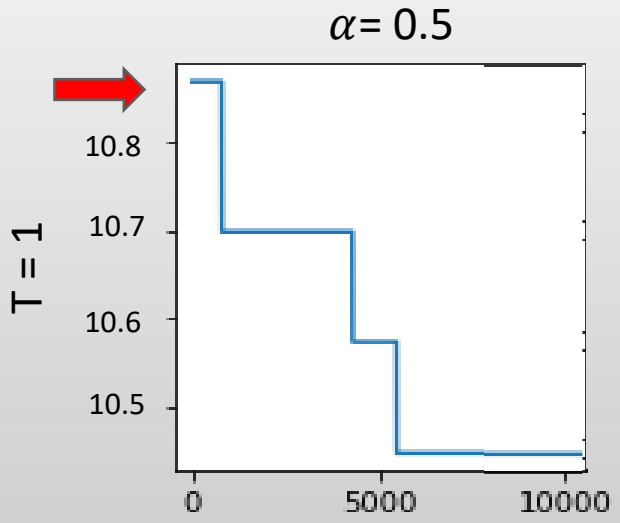
- Result for distance of different α and T with 10000 steps MCMC starting in local minimum (result of previous MCMC)



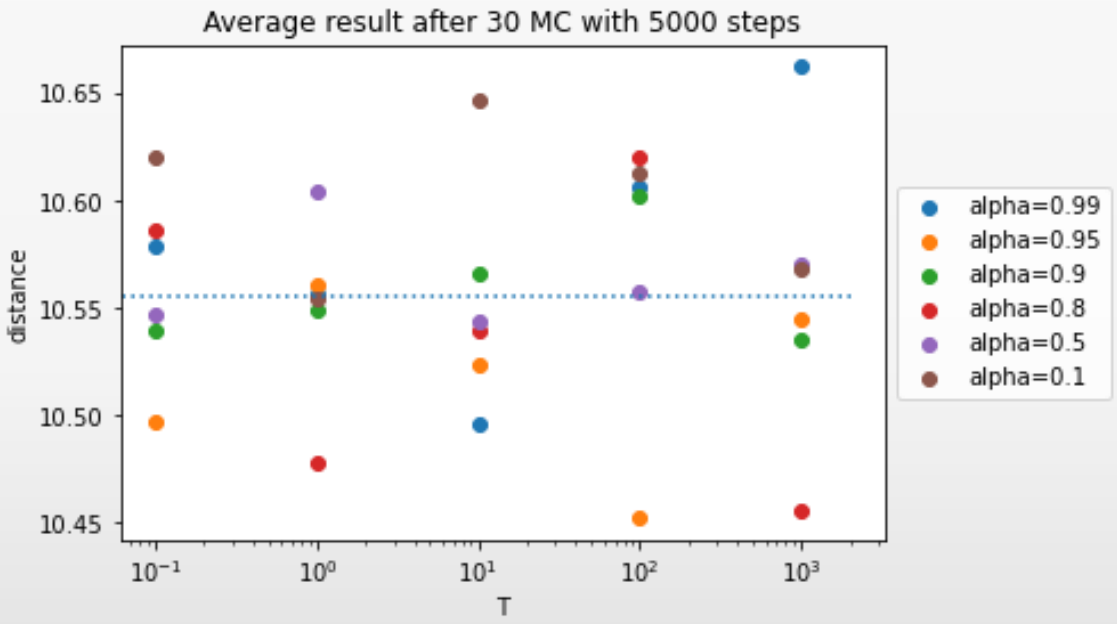


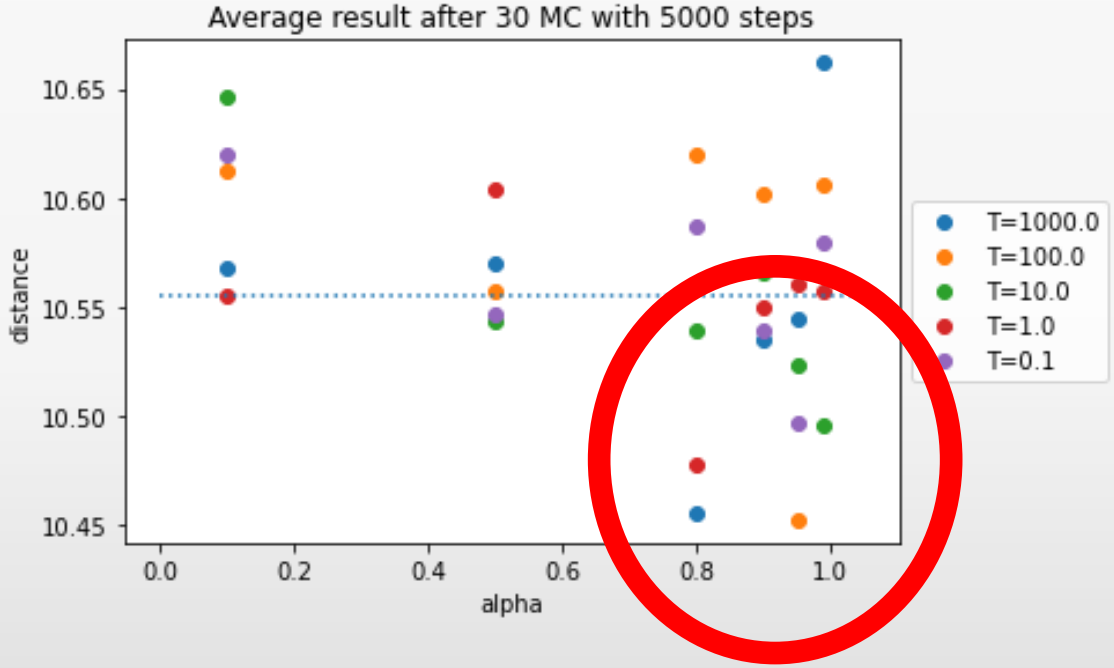






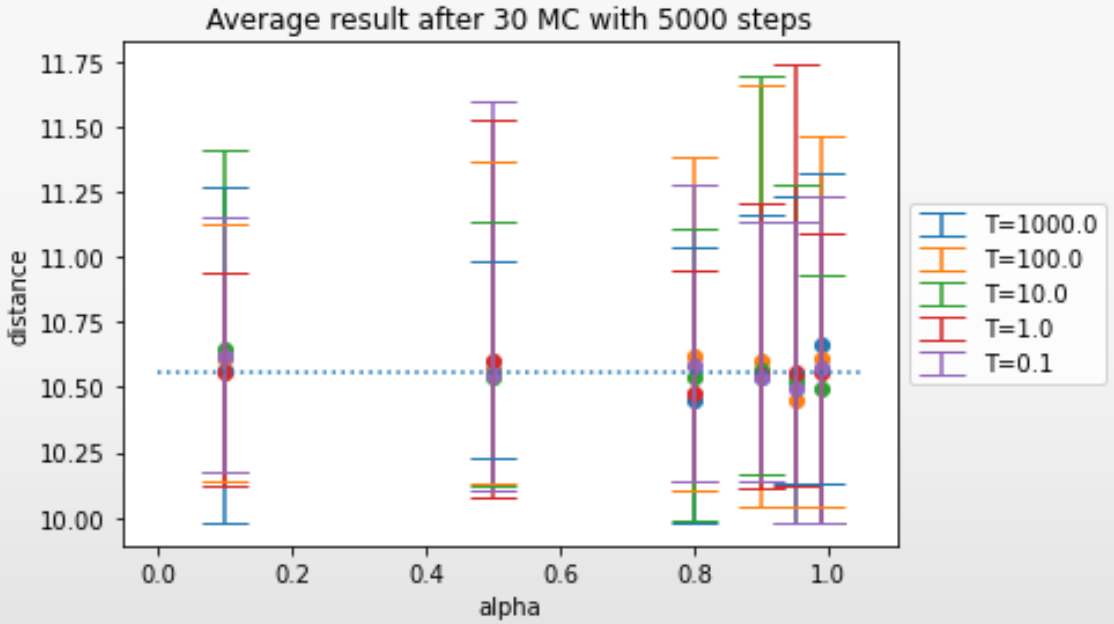
T and α averages over 30 MCMC

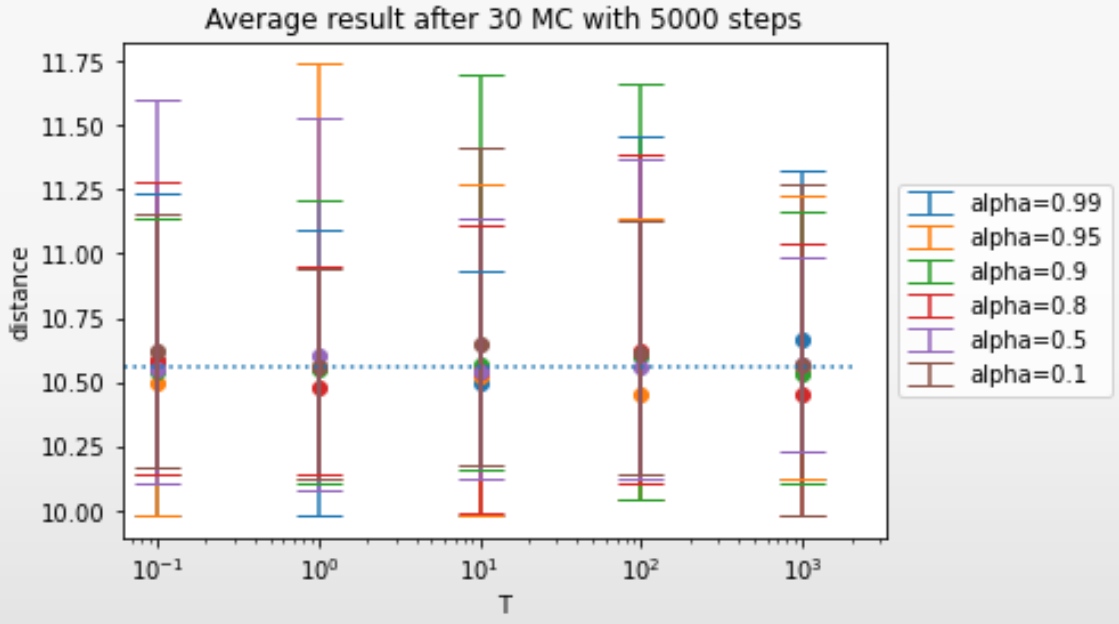




- In Average lower result for $\alpha > 0.8$

T and α spread of result over 30 MCMC

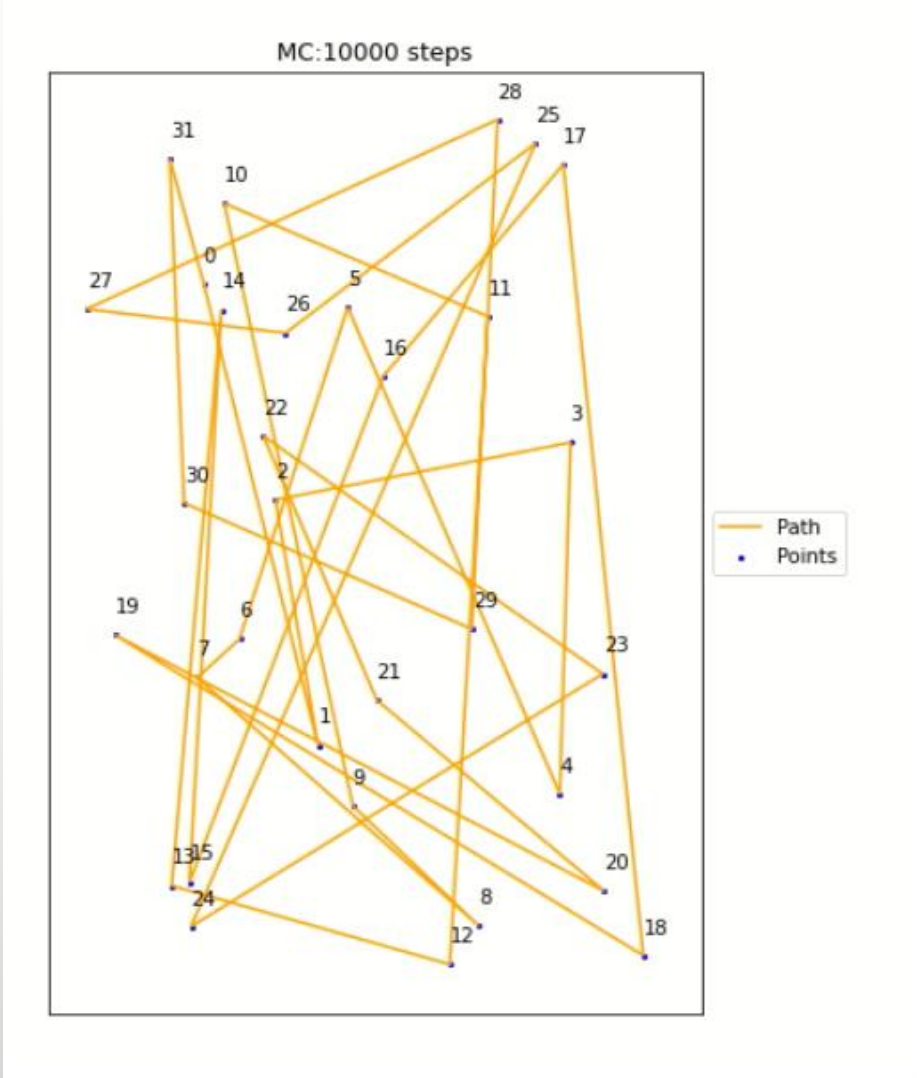




- Repeated MCMC leads to good result independently of chosen α and T due to randomness

Monte-Carlo - Animation

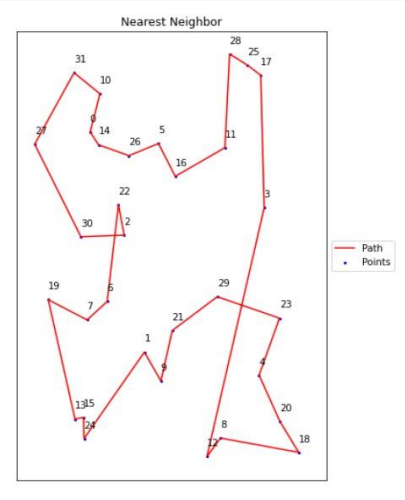
- Start with sequence 0,1,…,32
- Used 10000 steps
- T =100 and alpha = 0,95
- 20 frames per animation step



Comparison

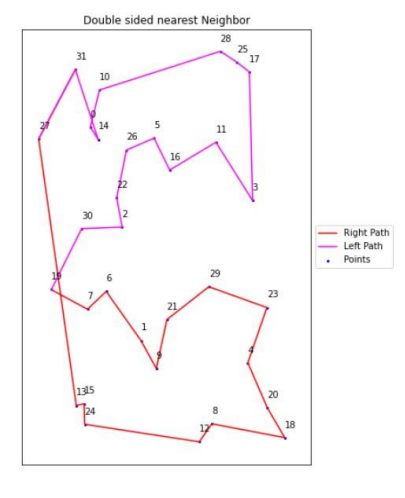
Comparison - Time

Nearest Neighbor



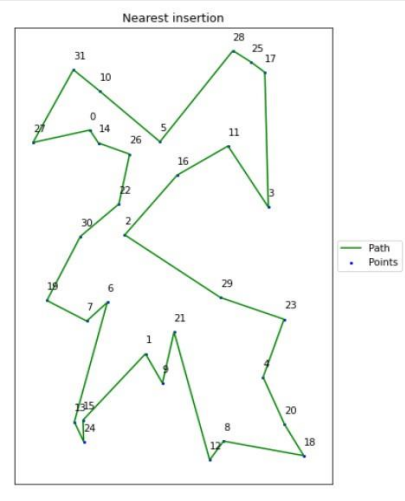
Time:
68.2 ms ± 2.63 ms

Double- sided nearest Neighbor



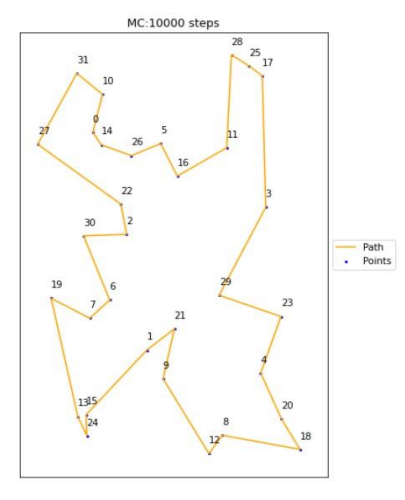
Time:
77.5 ms ± 5.04 ms

Nearest Insertion



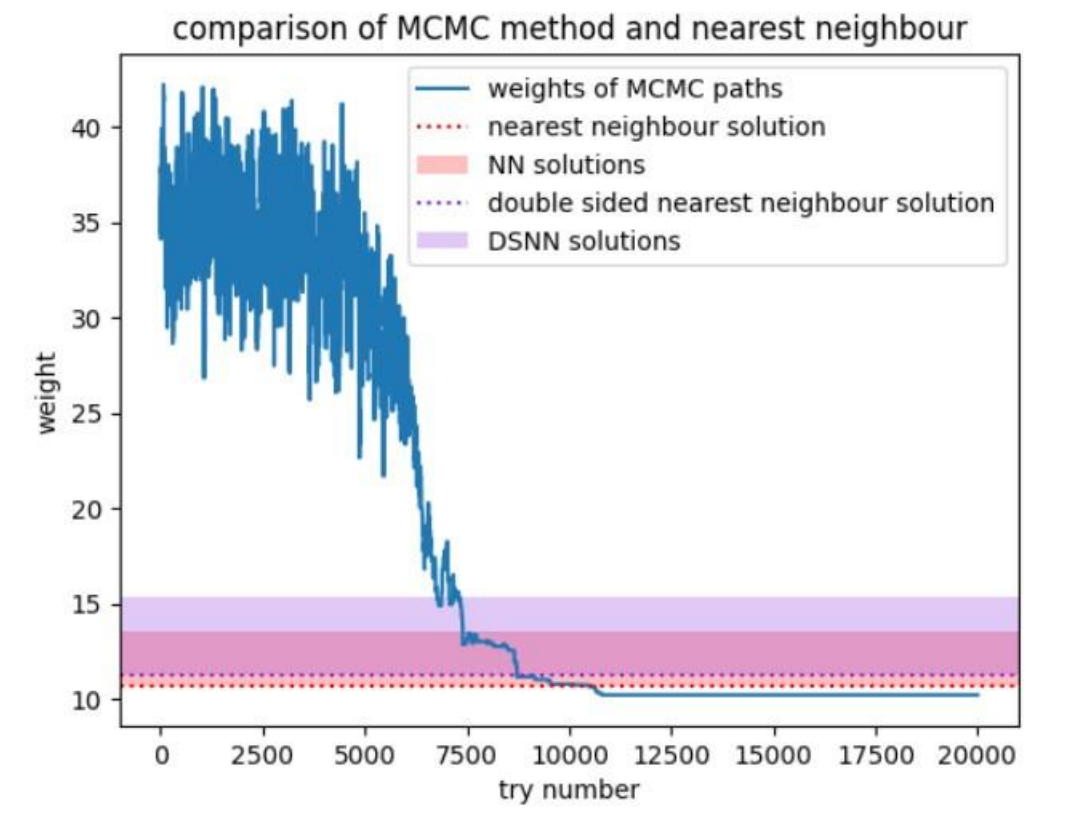
Time:
62 ms ± 1.11 ms

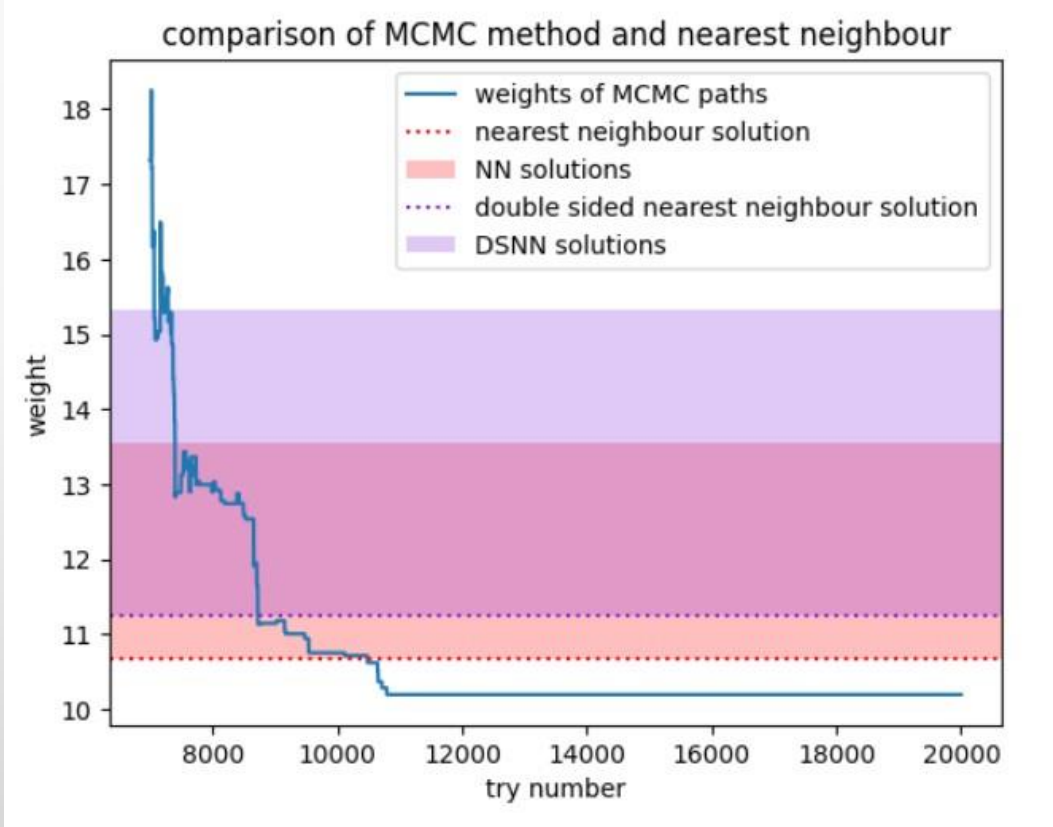
MCMC



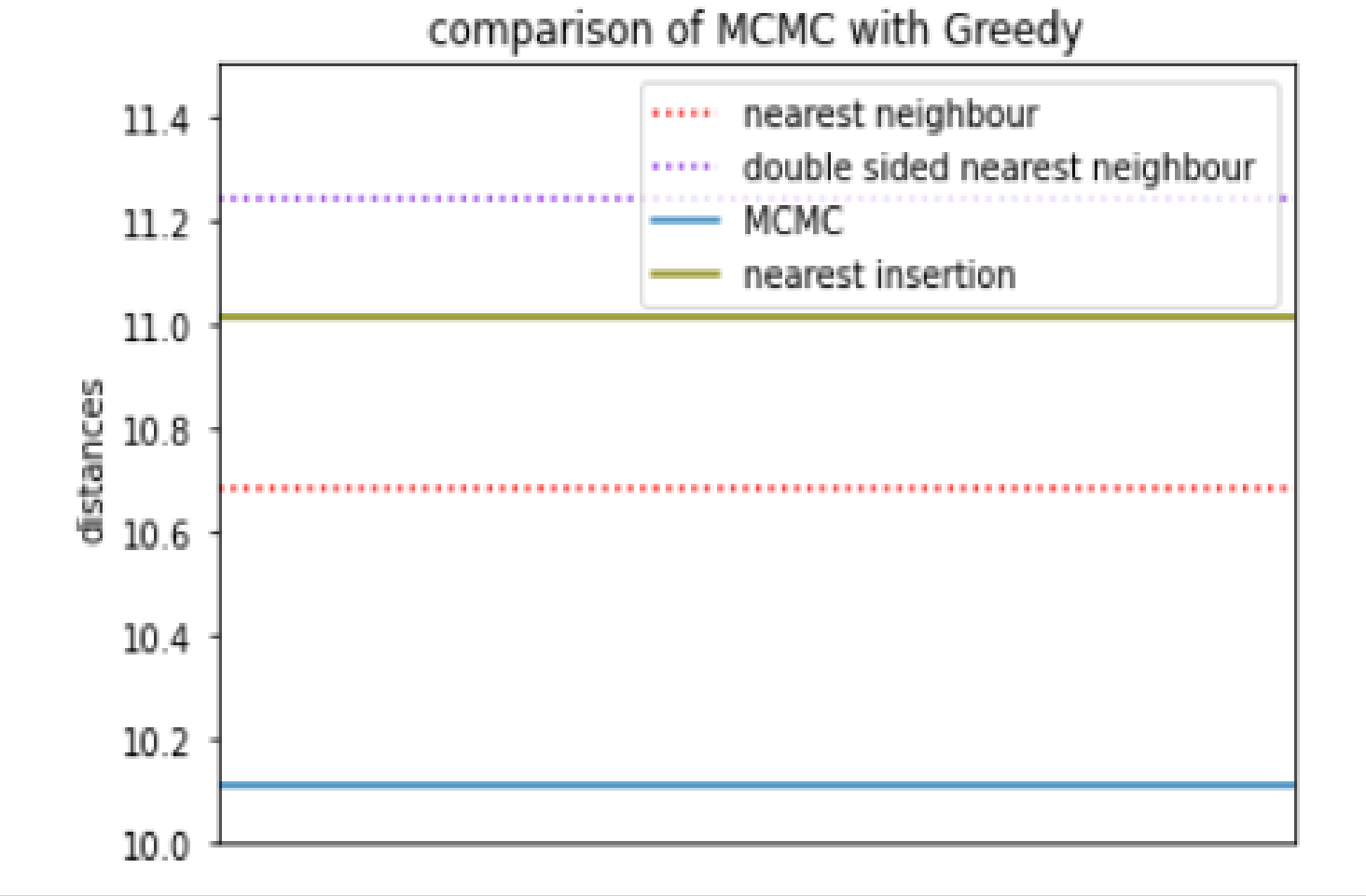
Time:
1.16 s ± 140 ms

Comparison - Distances





Comparison - Distances



Distance:11.243

Distance:11.012

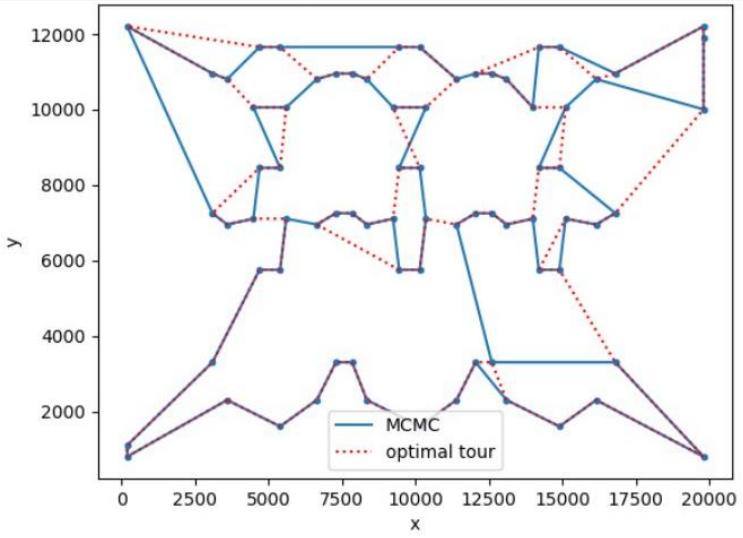
Distance:10.685

Distance:10.106

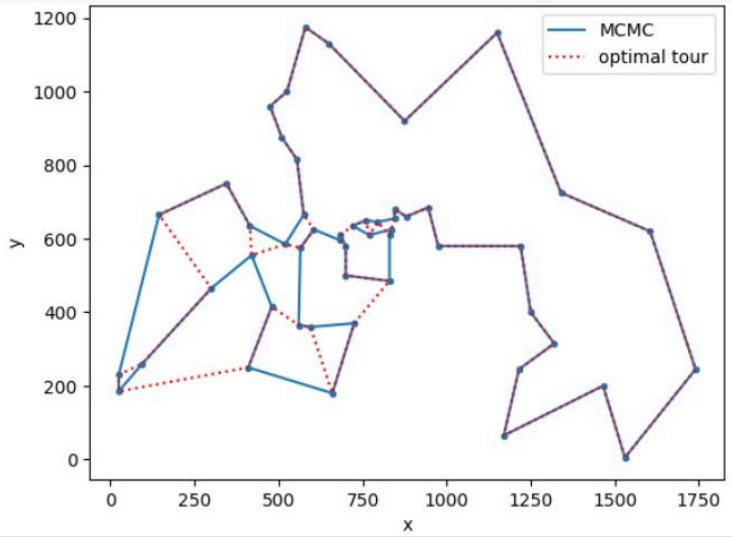
Comparison with known optimal solution

Optimal result from: http://comopt.ifi.uni-heidelberg.de/software/TSPLIB95/tsp/

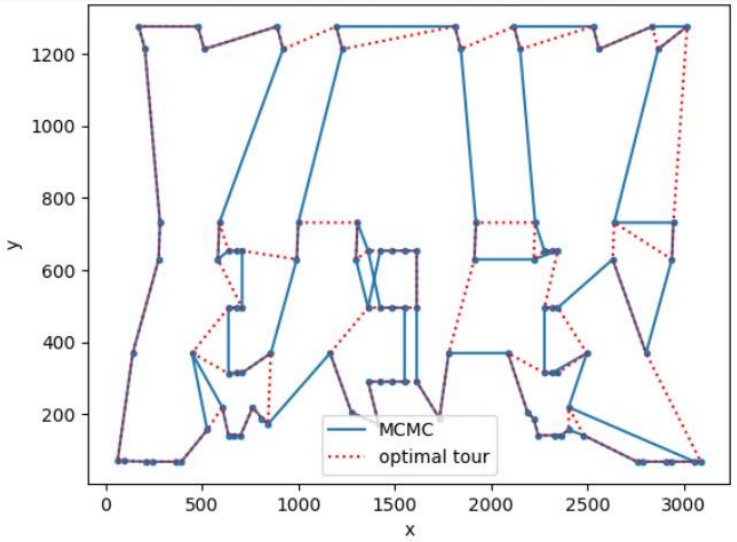
pr76



berlin52



lin105



Difference to optimal tour in %		
5,98	7,53	9,72

Optimization of the result

MCMC – Optimization of the result

- More steps -> Works until we land in local Minima
- As seen before: Multiple tries of MCMC
- Start in local Minima by reusing MCMC result
  - Do this a couple of time and keep best result:
    - For our example distance: 10.106 ->10.046 (50 tries)
- If group of points are clearly separated -> MCMC on subgroup and find cheapest connection between those

Further more complex Problems

Further more complex Problems

- Asymmetric distance Matrix
- Time dependent distance Matrix (e.g. traffic jam)
- m traveling salesmen
  - find best distribution of N cities for the m traveling salesmen and find shortest path over all $(N + m - 2)!$ possibilities

Thank you for your attention

(3)

Literature

- [1] “Double-ended nearest and loneliest neighbour – a nearest neighbour heuristic variation for the travelling salesman problem”, Fernando G. S. L. Pimentel https://www.researchgate.net/publication/258241208_Double-ended_nearest_and_loneliest_neighbour-a_nearest_neighbour_heuristic_variation_for_the_travelling_salesman_problem
- [2] „TRAVELING SALESMAN PROBLEM -Insertion Algorithms“ https://www2.isye.gatech.edu/~mgoetsch/cali/VEHICLE/TSP/TSP009__.HTM
- [3] “Optimization of the time-dependent traveling salesman problem with Monte-Carlo methods”,Bentner et al., Phys.Rev. E, 64, 036701

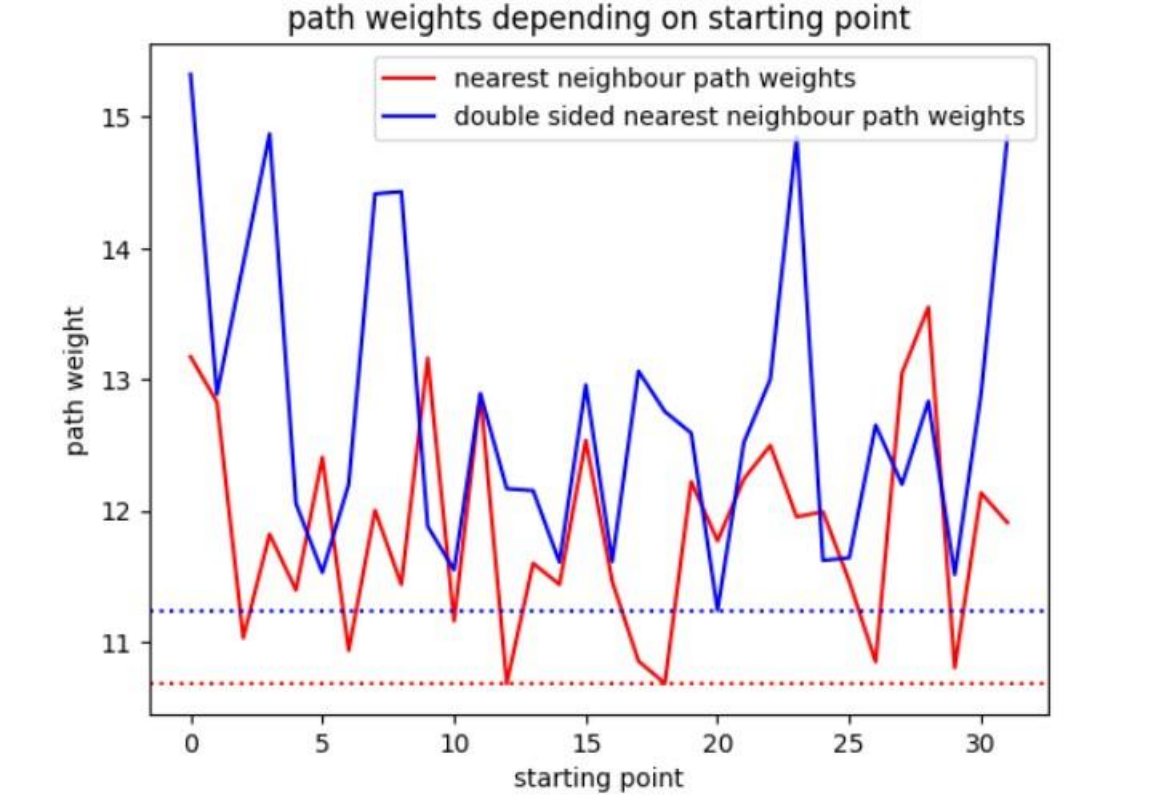
Source for images

- (1) https://www.tue.nl/en/news-and-events/news-overview/19-12-2023-improving-optimization-efficiency-by-modifying-algorithms-in-tsp-variants
- (2)https://www.indiskretionehrensache.de/2012/06/schufa-facebook/istock-manager-verwirrt-fragezeichen-tafel/
- (3)https://pxhere.com/de/photo/1446817

Back-up

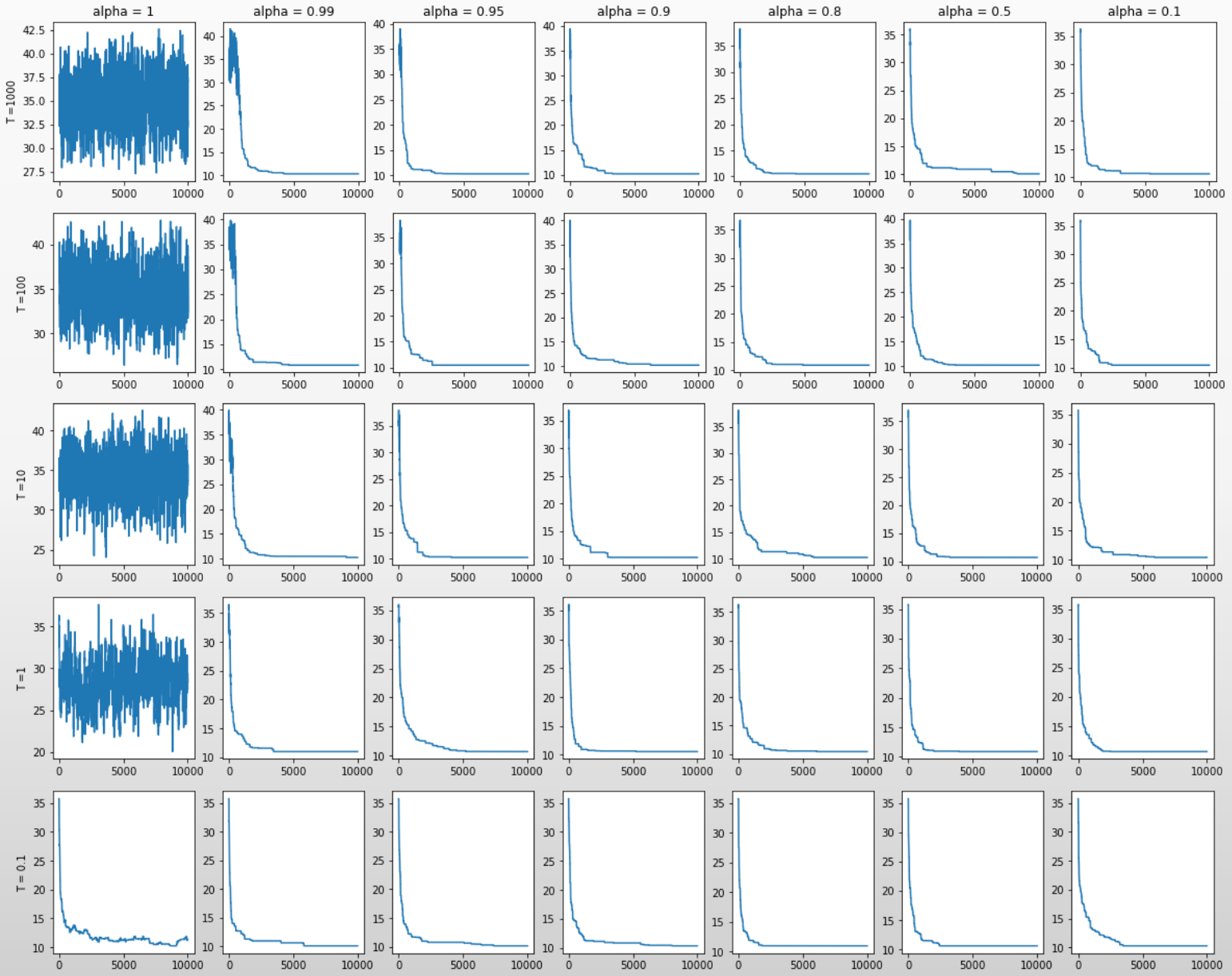
Comparison Nearest with double sided

- Does startposition matter? Yes

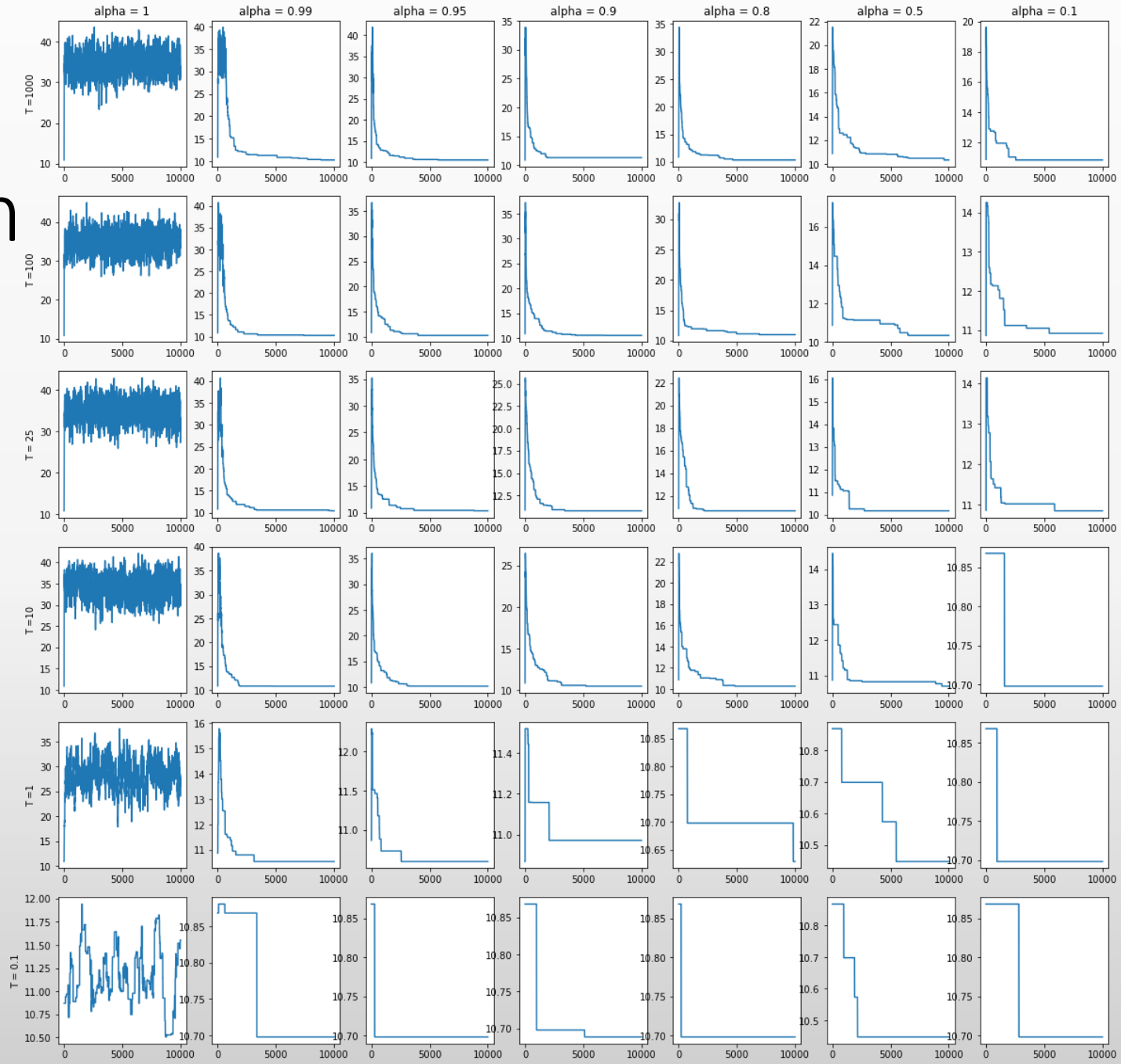


Different T and α 0,1,… start

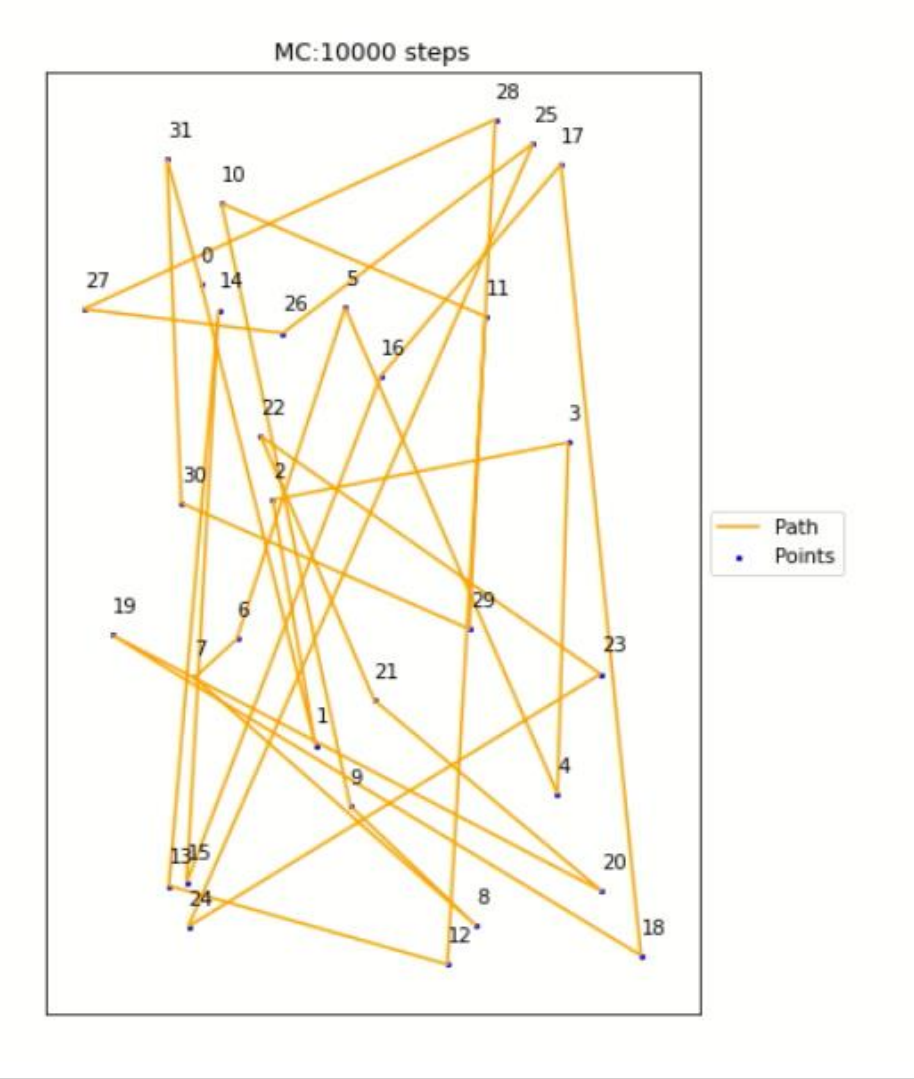
- Result for distance of different α and T with 10000 steps MCMC starting with 0,1,…,32
- We have $\exp\left(\frac{D-D^*}{T}\right)$ with $\max(D - D^*) \approx 2{,}5$

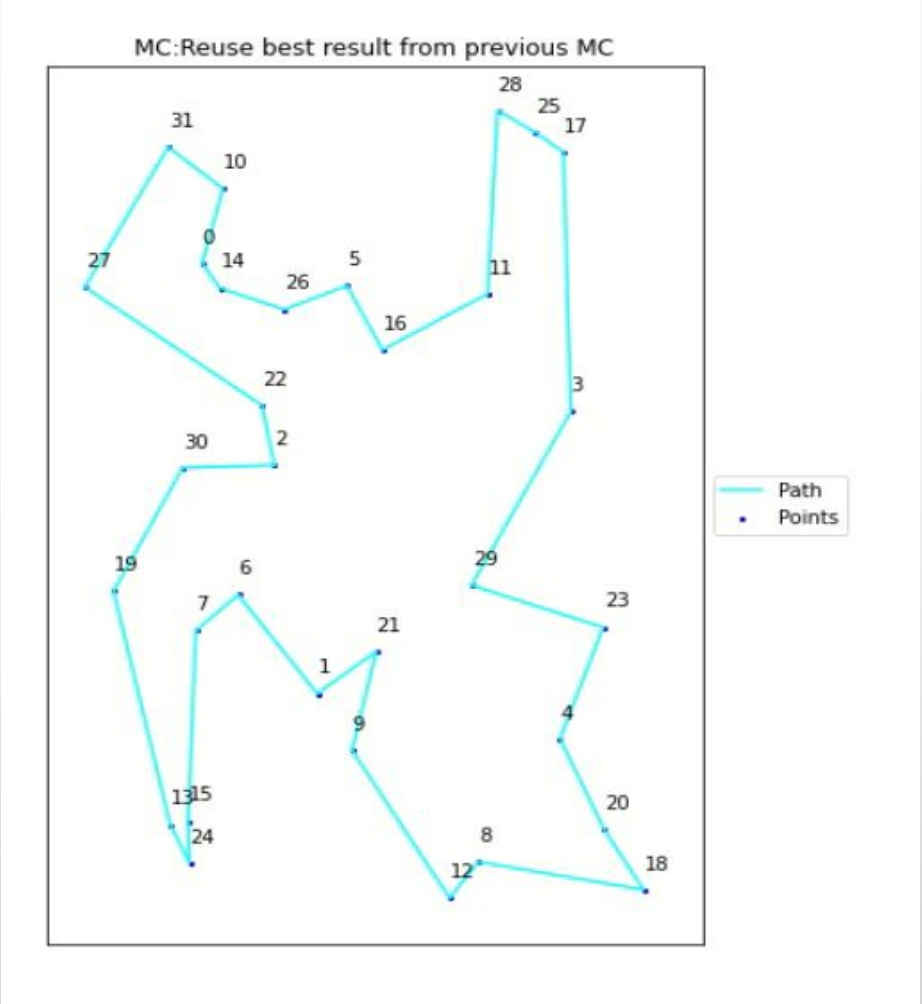


Different T and α start in local Minimum



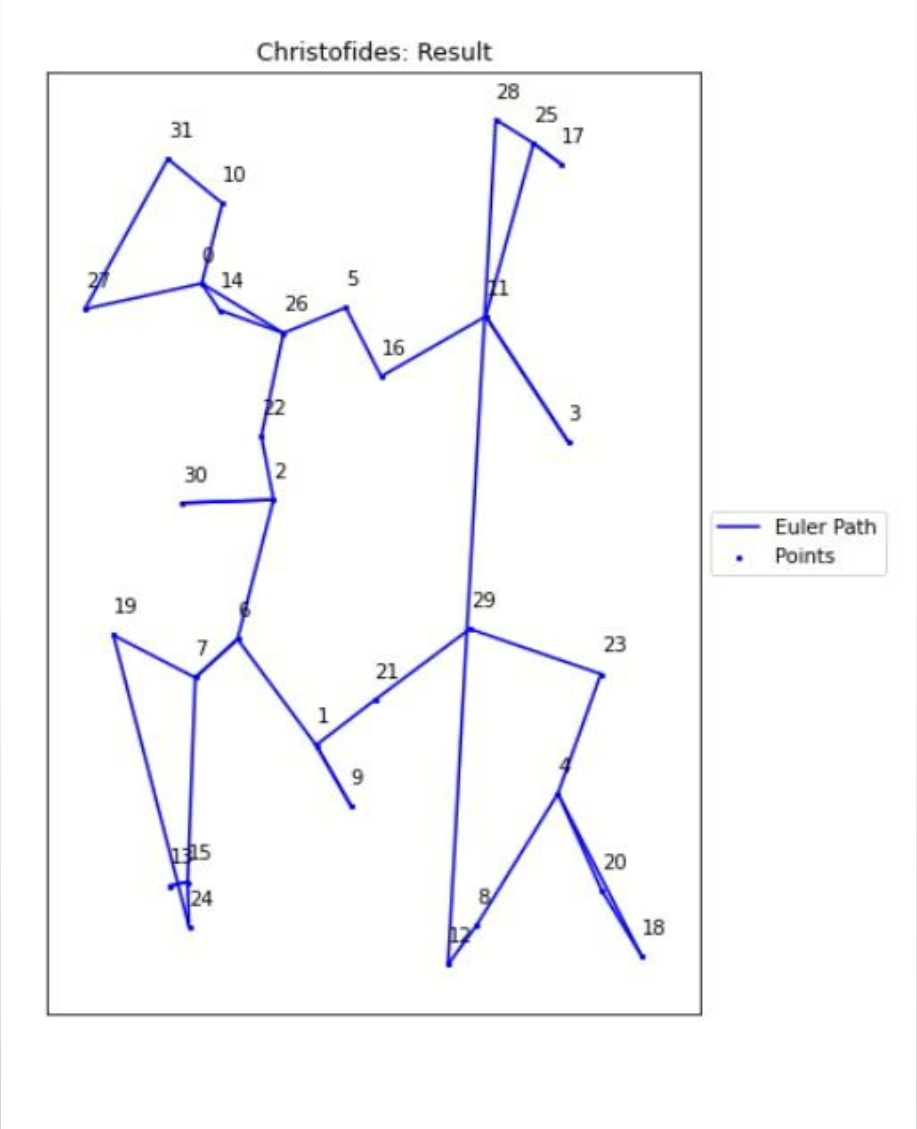
Time comparison local Minima/random start





Christofides

- Create a minimal spanning tree $\mathbb{T}$
- Find vertex with odd number of edges
- Find a perfect matching $\mathbb{M}$ between those vertex
- Create an eulerian circuit in $\mathbb{T} \cup \mathbb{M}$
- Create Hamiltonian circuit by skipping repeated vertices starting in a random point
- Minimize over all starting points



Code- Greedy

```
def Nearest_Neighbor(points,distance,start):
    """
        Calculate way with minial distance through Nearest_Neighbor algorithm with given start point
        Input: points - points of interests - integer array from 0-#cities-1
               distances - 2d-matrix containing the distances
               start- startpoint - integer
        Output: way with minimal_distances- by Nearest_Neighbor algorithm

    """
    way = np.array([start]) # start with startpoint

    while len(way) != len(points):
        min_dist = np.max(distance)+1

        for  i in range(len(points)):

            if (distance[way[-1],i] < min_dist) and (i not in way):
                # minimal distance and not yet visited city
                min_dist = distance[way[-1],i]
                new_point = i
        way = np.append(way,new_point)

    way = np.append(way,start) # end with startpoint
    return way
```

```
def Double_side_Nearest_Neighbor(points,distance,start):
    """
        Calculate way with minial distance through Double_side_Nearest_Neighbor algorithm with given start point:
        nearest neighbour from left and right.
        Input: points - points of interests - integer array from 0-#cities-1
               distances - 2d-matrix containing the distances
               start- startpoint - integer
        Output: way with minimal_distances- by Double_side_Nearest_Neighbor algorithm

    """
    way_right = np.array([start]) # start with startpoint
    way_left = np.array([start])
    used = np.array([start]) # used
    while (len(way_right)+len(way_left)-1) < len(points):
        min_dist = np.max(distance)+1

        for  i in range(len(points)):

            if (distance[way_right[-1],i] < min_dist) and (i not in used):
                # minimal distance and not yet visited city
                min_dist = distance[way_right[-1],i]
                new_point = i
        way_right = np.append(way_right,new_point)
        used = np.append(used,new_point)
        min_dist = np.max(distance)+1
        if len(used) != len(points):
            for  i in range(len(points)):
                if (distance[way_left[0],i] < min_dist) and (i not in used):
                    # minimal distance and not yet visited city
                    min_dist = distance[way_left[0],i]
                    new_point = i
            way_left = np.append(new_point,way_left)
            used = np.append(used,new_point)
    way = np.append(way_right,way_left) # end with startpoint
    return way
```

```python
def nearest_Insertion(distance):
    """
        Implementation of nearest Insertion method: inserting nodes to cycle, so that length increase is minimal
        Input: distance 2d-matrix
        Output: samllest path regarding nearest Insertion method

    """

    N = len(distance)
    points = np.arange(N)
    x = np.random.choice(points, size=1, replace=False)
    minn = np.max(distance)
    for i in range(len(points)):
        if i!=x:
            if distance[i,x]<minn:
                minn = distance[i,x]
                y =i
    cycle = np.append(x,y)
    while len(cycle) <N:
        d = np.max(distance)
        for c in range(len(cycle)-1):
            pos1 = cycle[c]
            pos2 = cycle[c+1]
            for j in range(N):
                if j not in cycle:
                    if distance[pos1,j]+distance[j,pos2]-distance[pos1,pos2] <d:
                        d = distance[pos1,j]+distance[j,pos2]-distance[pos1,pos2]
                        k = j
                        rep_c = c+1
        cycle = np.insert(cycle,rep_c,k)


    cycle = np.append(cycle,cycle[0])
    return cycle
```

Code- MCMC

```python
def twoPointDist(x, y, dist_matrix):
    """ input:  two points x and y - (int)
                matrix giving the distances between points - (2d array)
        return: distance (int) between point x and point y (direction x to y)
    """
    return(dist_matrix[x, y])

def totalTravelDist(points, dist_matrix, loop=True):
    """ input:  points - array of points that gives the path - array
                dist_matrix - matrix containing the distances between all points (in both directions) - 2d array
                loop - info if the path needs to be a loop (start position = end position) - boolean
        output: dist_sum - total weight of the given path - int
    """
    N = len(points)
    if loop:
        dist_sum = twoPointDist(points[N-1], points[0], dist_matrix)
        for i in range(N-1):
            dist_sum += twoPointDist(points[i], points[i+1], dist_matrix)

    else:
        dist_sum = 0
        for i in range(N-1):
            dist_sum += twoPointDist(points[i], points[i+1], dist_matrix)
    return(dist_sum)
```

```python
def travelWeightTotal_changingDist(points, distances, periodicity, loop = True):

    """ input:  points - array of points representing the path - array
                distances - matrices containing the distances between all points (in both directions) at different 'times' (3rd dim = time
dim) - 3d array
                periodicity - periodicity of the time matrices in time - int
                loop - info if the path needs to be a loop (start position = end position) - boolean
        return: dist_sum - total weight of the given path - int
    """
    N = len(points)
    dist_sum = 0
    for i in range(N-1):
        dist_sum += twoPointDist(points[i], points[i+1], distances[round(dist_sum) % periodicity,:,:])

    if loop:
        dist_sum += twoPointDist(points[N-1], points[0], distances[round(dist_sum) % periodicity,:,:])
    return(dist_sum)

def exchangeTwoPartsorder(x1, x2, y1, y2, arr):
    """ input:  x1, x2, y1, y2 - positions in array at which the array is to be cut in subarrays (it needs to be x1<x2<y1<y2) - int
                array - array which is to be reordered - array
        return: reordered array (exchange the position [0,...,x1,...,x2,....,y1,...y2,...] to [0,...,y1,...,y2,....,x1,...x2,...]) - array
    """
    return(np.concatenate((arr[0:x1],arr[y1:y2+1], arr[x2+1:y1], arr[x1:x2+1], arr[y2+1:])))

def Lin_3_Opt (path):
    """ input:  path - array that is to be reordered - array
        return: reordered path (exchanges two random successive parts of the tour without altering their directions) - array
    """
    ordered_cuts = np.array([0,0,0,0])
    while ordered_cuts[1]== ordered_cuts[2]:

        cuts = np.random.randint(len(path), size=4) ##take for random positions in path-array and put them in array
        ordered_cuts = np.sort(cuts)                #sort the array from smallest to largest

    return(exchangeTwoPartsorder(ordered_cuts[0],ordered_cuts[1],ordered_cuts[2],ordered_cuts[3],path))
```

```python
def Lin_2_Opt(path):
    """ input:  path - array that is to be reordered - array
        return: reordered path (Lin-2-Opt move simply reverses the direction of a part of the tour ) - array
    """
    positions = np.arange(len(path))

    cut =  np.random.randint(len(path),size =2) ##take random postion in path-array
    order_cut = np.sort(cut)
    change = np.concatenate((positions[0:order_cut[0]], positions[order_cut[0]:order_cut[1]+1][::-1], positions[order_cut[1]+1:]))
    ##make the change with the already ordered positions
    return(path[change])

def Opt_method(x,array):
    """
    Chooses method of path change

    input: x - = 0,1 or 2: 0: Only Lin-2, 1: Only Lin-3, 2: randomize between both
        array - array to be alterad
    output: new array with either Lin-2 or Lin-3 used upon
    """
    if x == 0:
        return Lin_2_Opt(array)
    elif x == 1:
        return Lin_3_Opt(array)
    else:
        method = np.random.choice(2) # choose if you use Lin
        if method ==1:
            return Lin_2_Opt(array)
        else:
            return Lin_3_Opt(array)
```

```python
def traveling_MC_constDist(points, dist_matrix, tries, starting_temp = 1, alpha = 0.999, method =2):
    """ input:  points - points that should be visited by each path (in some order) - array
                dist_matrix - matrix containing the 'distances' between each point - 2darray
                tries - itterations of the Monte Carlos - Markov chain - int
                starting_temp - scalar in the exponent of the markov chain acception process - int
                alpha - scalar refgulating the decrease of the temperature after each loop iteration in the MC
                method - see Opt_method- cuurently random between Lin-2 & Lin-3
        return: points - points that should be visited by each path (in some order) - array
                pathWeights - 'distances' (better weights) of each path - array of int
                temps - temperatures after each MC step - array of int
    """
    points = np.append(points,points[0])
    pathWeights = np.zeros(tries+1)
    temps = np.zeros(tries+1)
    pathWeights[0] = totalTravelDist(points, dist_matrix)
    temps[0] = starting_temp
    cost = pathWeights[0]

    for i in range(tries):
        points = np.delete(points,-1)
        new_sequence = Opt_method(method,points)
        new_sequence = np.append(new_sequence,new_sequence[0])
        new_cost = totalTravelDist(new_sequence, dist_matrix)
        if np.random.uniform(low=0, high=1) < np.exp((pathWeights[i]-new_cost)/temps[i]):
            points = new_sequence
            cost = new_cost
        else:
            points = np.append(points,points[0])
        pathWeights[i+1] = cost
        temps[i+1] = alpha*temps[i]
    return(points, pathWeights, temps)
```